



lushprotein

Lush Protein

Final Presentation – **ISSS603**

Group 1



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Project Overview

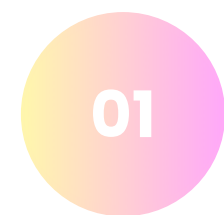
From “**low retention**” to a **measurable customer growth system**

THE PROBLEM, RESTATED

*Most LushProtein customers never build a repeat habit in the first place — **only 11% reach 3+ orders** — and **once that habit fails to form early, they go quiet** – **49% of our highest-margin customers haven’t ordered in over a year.***

TWO SOLUTIONS, ONE GOAL

Move customers up the value ladder — and bring back the ones we’re losing



Customer Treatment (Retain + Win)

Turning customer intelligence into reinvestment action

Decile and tier the existing base by **profit, frequency**, and **recency** — then match every segment to a **concrete, non-discount campaign**. **Protects** who we have and **wins back** who we’re losing.

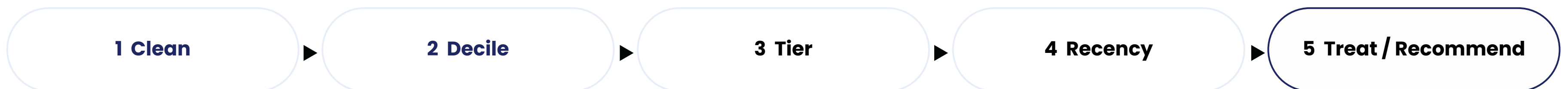


Recommender System (Develop)

Developing lower-decile customers so they move up

Match the **right algorithm to the right customer** moment — first purchase, active browsing, post-purchase, repeat buyer — instead of one model for everyone. **Builds the habit that’s currently missing.**

HOW WE GET THERE



Profit Decile & AOF – from 13,780 raw customers → 4,290 real, rankable consumers

PREPARING FOR DECILING

13,780

All customers ever recorded

6,353

Excl. pre-2022, promo-month, bulk-SKU, heavy-discount

4,949

Excl. 100% Shopee/Lazada (resellers)

4,290

Final pool – real, rankable consumers

68.9% of customers are excluded on purpose – pre-2022 cohorts, promo-driven one-timers, and marketplace resellers don't reflect real, **individually-rankable consumer behaviour**.

DECILE 1 – PROFIT (Revenue - COGS)

Decile	Customers	Avg Profit	Avg Revenue	Avg Orders	Margin %	% of Profit
D1	858	S\$230.05	S\$374.69	3.1	69.1%	57.7%
D2	858	S\$81.25	S\$122.04	1.5	70.5%	20.4%
D3	858	S\$48.59	S\$74.48	1.2	69.3%	12.2%
D4	858	S\$29.66	S\$52.75	1.1	60.7%	7.4%
D5	858	S\$8.87	S\$19.25	1.1	49.3%	2.2%

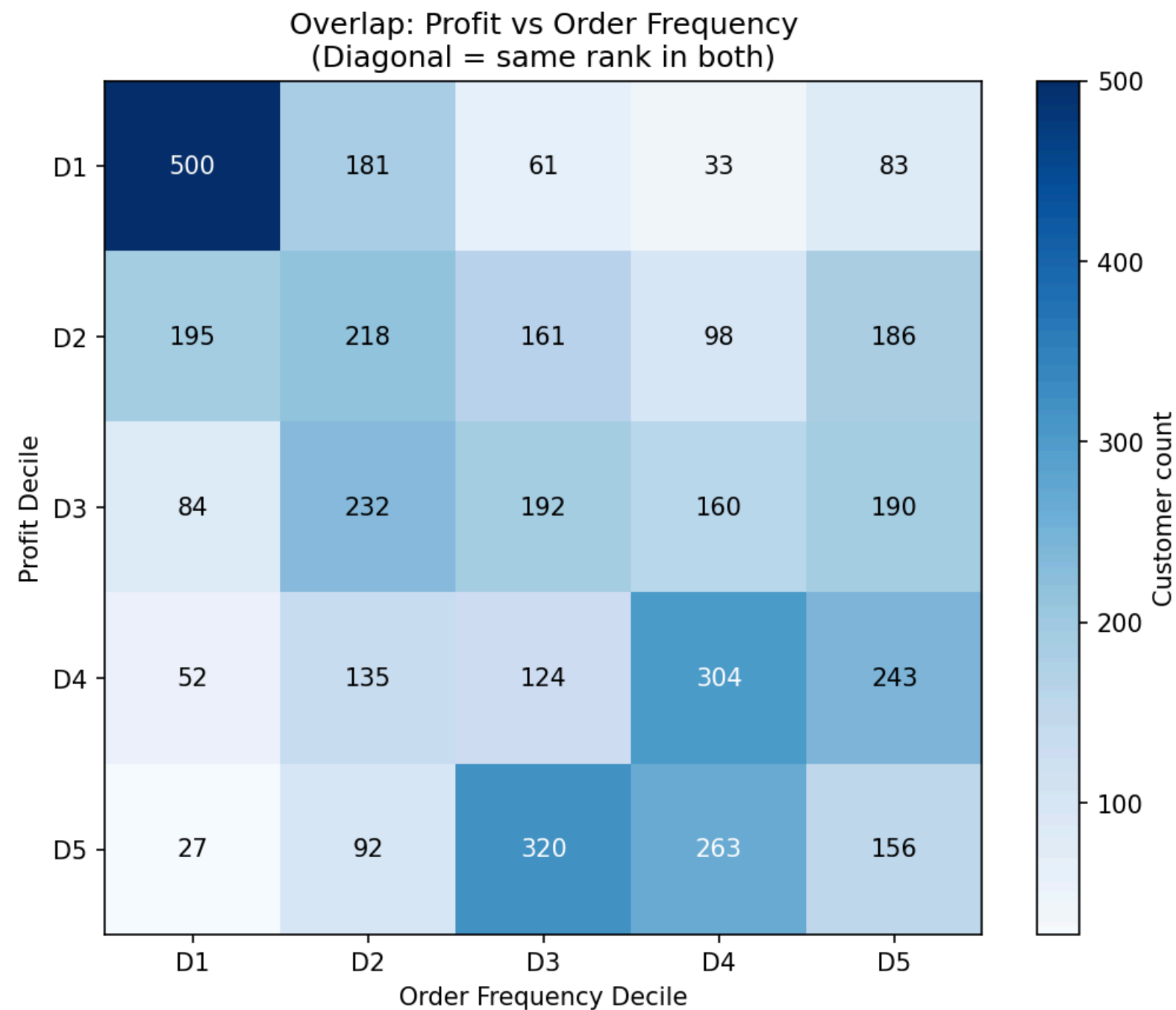
DECILE 2 – ORDER FREQUENCY

Decile	Customers	Avg Orders	Avg Revenue	Avg Profit	Margin %	% of Orders
D1	858	3.6	S\$297.30	S\$180.60	62.9%	45.4%
D2	858	1.3	S\$125.77	S\$80.31	68.9%	16.6%
D3	858	1.0	S\$58.75	S\$42.75	67.6%	12.6%
D4	858	1.0	S\$68.19	S\$38.99	56.6%	12.6%
D5	858	1.0	S\$93.22	S\$55.78	62.8%	12.6%

Why two deciles? A customer can **spend a lot once**, or **a little many times** – both signal value **differently**. D1 in profit and D1 in frequency are not always the same people.

Tier

Forming 4 different tiers using the deciles



Profit Decile × Frequency Decile overlap

TIER 1 (PREMIUM)

D1 Both

Highest **profit** AND highest **frequency** — the truly best

500 customers

TIER 2

D1 Profit Only

High **spenders**, but not necessarily loyal

358 customers

TIER 3

D1 Frequency Only

Loyal habit, but low-margin baskets

358 customers

TIER 4

All D2 (excl. above)

Strong **second** tier, not yet top-decile in either

1,122 customers

Tiers are mutually exclusive — a Tier 1 customer is not double-counted in Tier 2 or 3.

Solution 1 – Customer Treatment



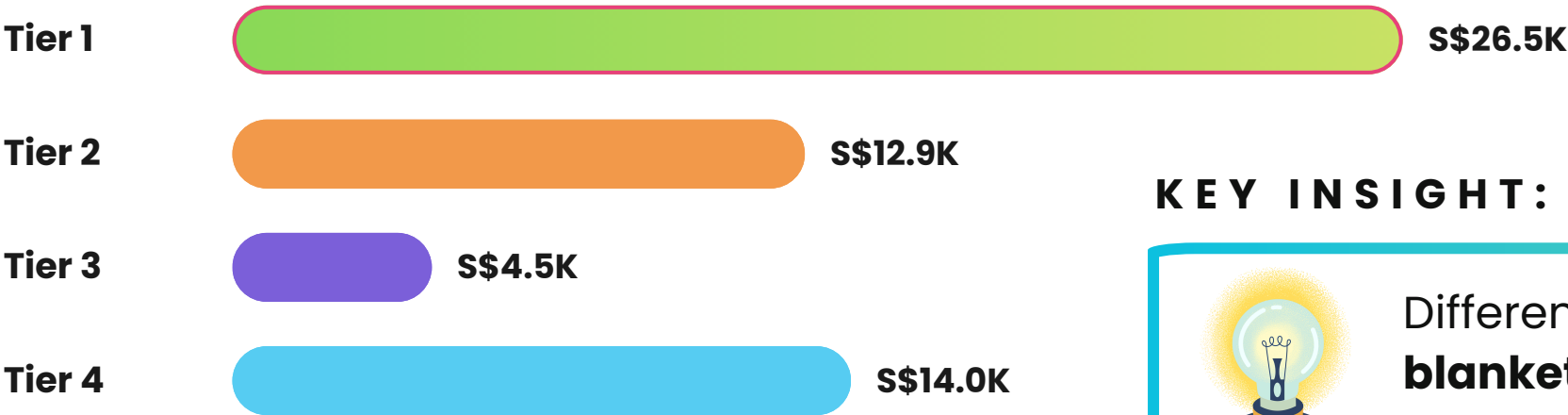
Turning customer intelligence into reinvestment action.

Tier	#	Active	At Risk	Lapsed	Profit	Reinv. 20%
T1 (D1 Both)	500	128	189	183	S\$132.7K	S\$26.5K
T2 (Profit Only)	358	45	81	232	S\$64.7K	S\$12.9K
T3 (Freq Only)	358	43	109	206	S\$22.3K	S\$4.5K
T4 (D2 Profit/Freq)	1,122	384	322	416	S\$69.9K	S\$14.0K

Note: Active (Last 3 months), At Risk (3–12 months), Lapsed (Over a year)

49% of the S\$57.9K budget sits with customers lapsed 12+ months — they need **win-back**, not loyalty rewards.

REINVESTABLE BUDGET BY TIER



KEY INSIGHT:



Different customers go quiet for **different reasons**. So they get **different treatment**, not a **blanket discount**. Each segment's campaign is funded by **20% of the profit that segment already generated**.

Segment	Cust.	Reinv.	Incentive Suggestions
T1 – Active	128	S\$7.4K	VIP early access + free full-size sample on launch
T1 – At Risk	189	S\$8.6K	Personal check-in + free travel-size gift
T1 – Lapsed, repeat	87	S\$6.4K	Founder outreach + free bundle of top SKU
T1 – Lapsed, tried	96	S\$4.2K	Automated 3-email win-back + free sample
T2 – Active+At Risk	126	S\$3.5K	Free 1-month subscription trial
T2 – Lapsed	232	S\$9.4K	Low-touch automated win-back only
T3 – Active+At Risk	152	S\$2.0K	Free sample of higher-margin adjacent SKU
T3 – Lapsed	206	S\$2.5K	Check for accidental subscription cancels
T4 – Active	384	S\$4.0K	Post-purchase nurture + free shaker on 2nd order
T4 – At Risk+Lapsed	738	S\$10.0K	Lowest priority – minimal automated email

Solution 1 – Customer Treatment

Implementation & Outcome

OUR SUGGESTION – Designed to fit each tier's actual reinvestable budget

PROTECT

Tier 1 Active + At Risk

317 Customers

Budget: **\$15,967** – **\$50.37/cust**

“VIP Pass” – Cover one full restock/subscription cycle (~\$50 retail value) for customers who haven't committed to a subscription yet.

With **57%** having subscribed before, the key barrier is **commitment**. A free cycle **encourages** long-term subscription behavior.

Cycle-1 spend: **\$3,804** (**30% x \$40**)

Gain: **\$6,248**, **+\$2,444**, **1.6X**

WIN BACK

Tier 1 Lapsed Repeat Buyers

87 Customers (4+ orders)

Budget: **\$6,403** – **\$73.60/cust**

Founder-personal outreach + a full dize bundle of their historical top SKU (~\$70 retail value)

They averaged **6 orders** before going quiet.

Cycle-1 spend: **\$3,480** (**\$40 x 87**)

Gain: **\$1,950** (**\$65 x 30**), **-\$1,530**

BUILD

Tier 4 Active

384 Customers

Budget: **\$4,022** – **\$10.47/cust**

Free shaker/ sample timed to land right at their natural reader window (~day 30)

Only **22%** have ever subscribed – not loyal yet.

Cycle-1 spend: **\$2,304** (**\$6 x 384**)

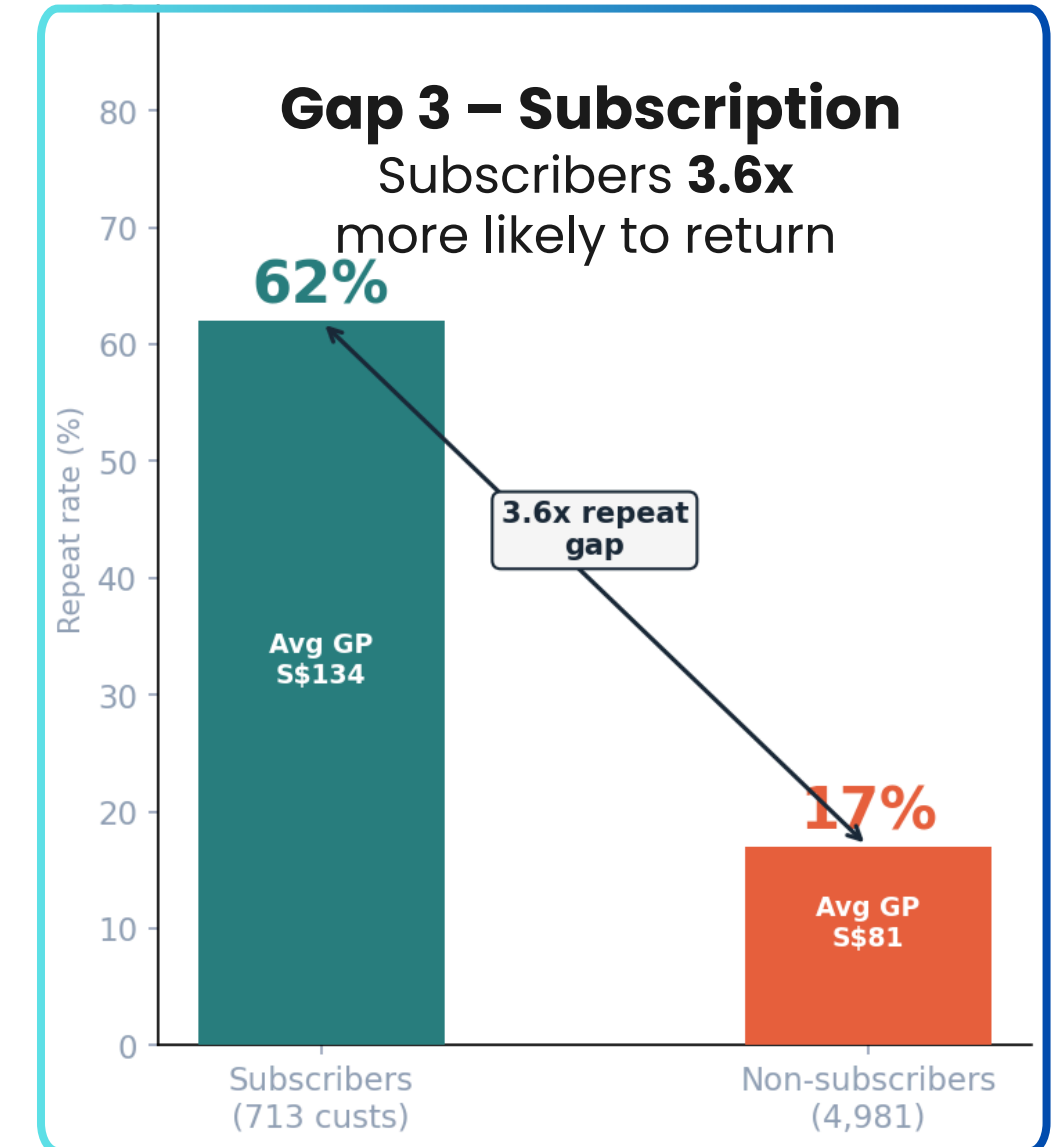
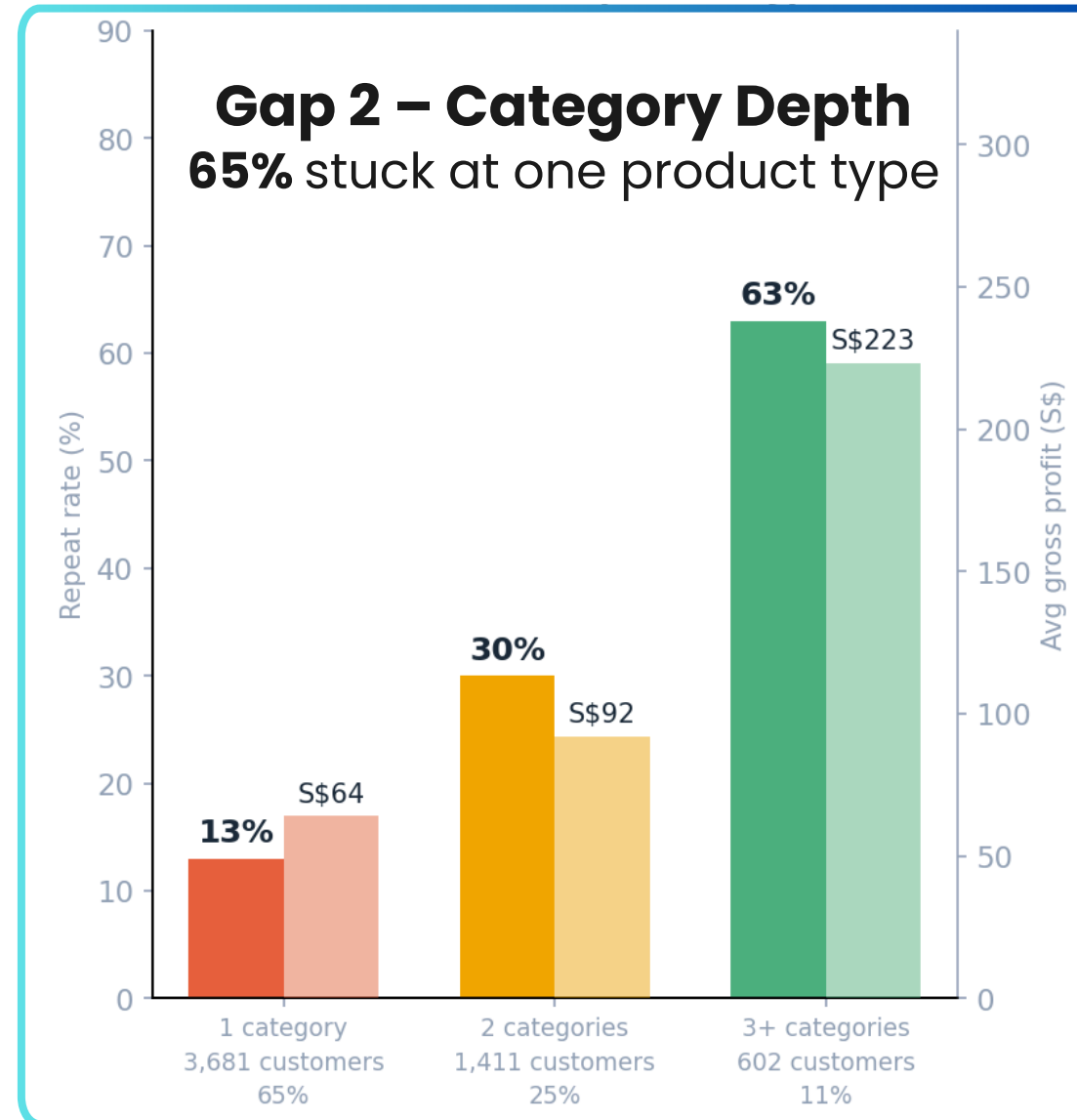
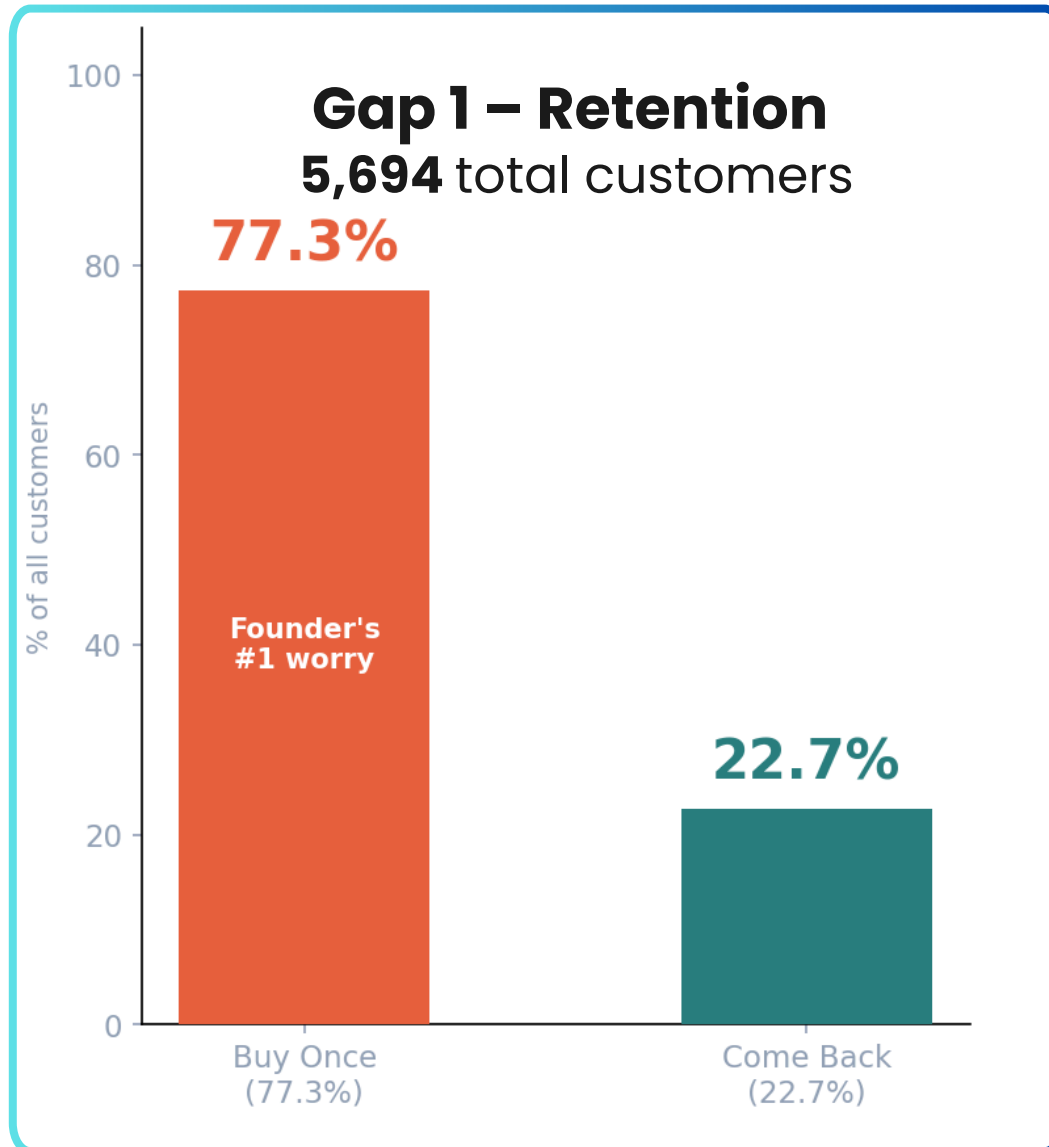
Gain: **\$4,413** (**22% x \$52.24**), **+\$2,109**, **1.9X**

BLENDED FIRST-CYCLE OUTCOME (3 priorities combined)

\$9,588 spent of **\$26,392** budget → **\$12,611** profit gained → **+\$3,023** net profit (**1.32x ROI**) – remaining **64%** of budget held in reserve for cycle 2

Solution 2 – Recommender System

Why we need it.



What's already working:

- D1 customers (**top 10%**) generate **57.7%** of total GP
- **287** VIP customers with **S\$476** average GP
- Clear Protein Peach 500g & Lean TMT 1kg drive **27–35% repeat rates**
- Collagen-first buyers repeat at **30% vs 18%** for accessories-first
- **65%** of D1 Lean buyers **also buy Clear, showing strong cross-sell demand**

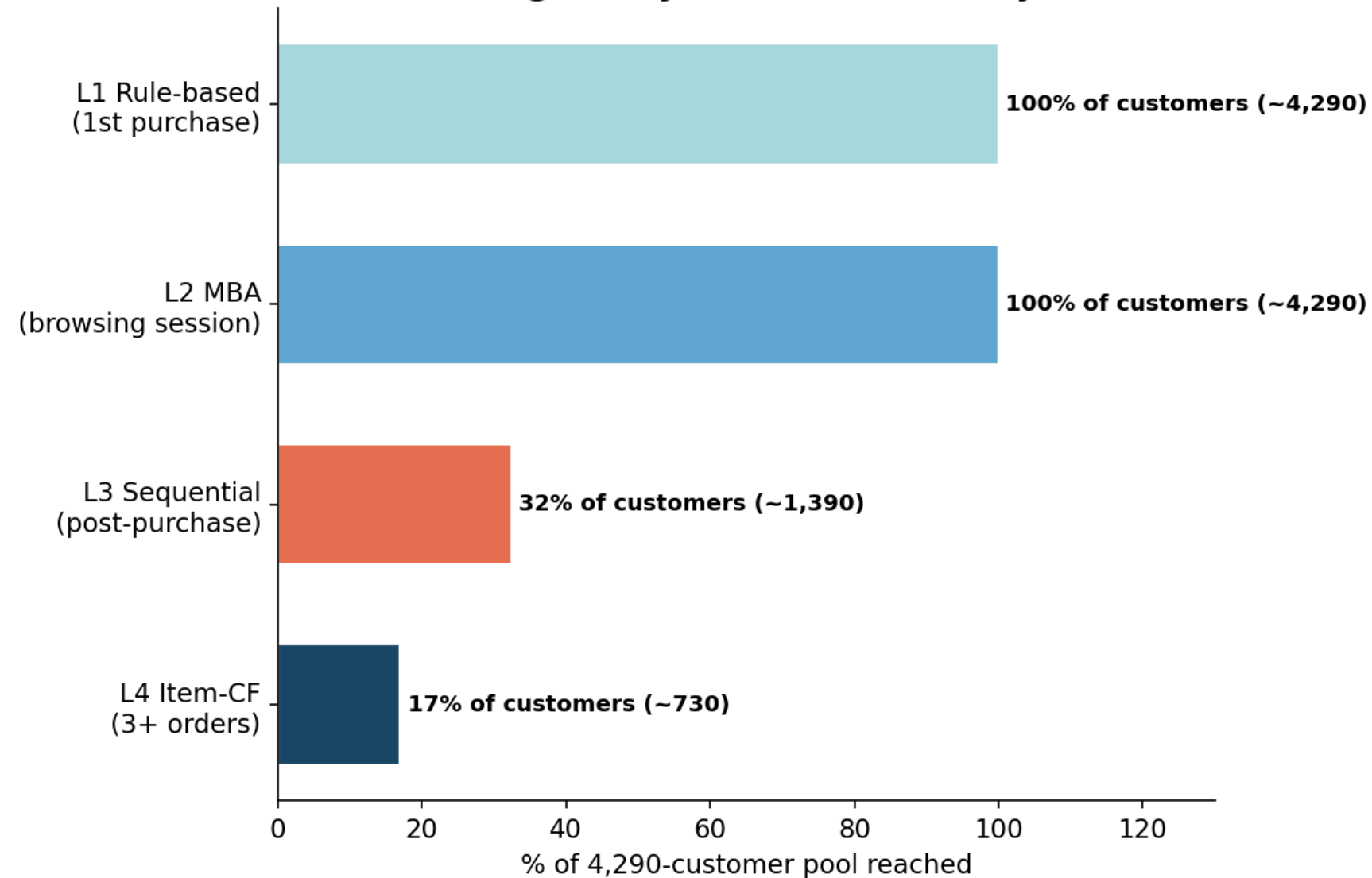


Recommendation system is built around these proven D1 customer traits.

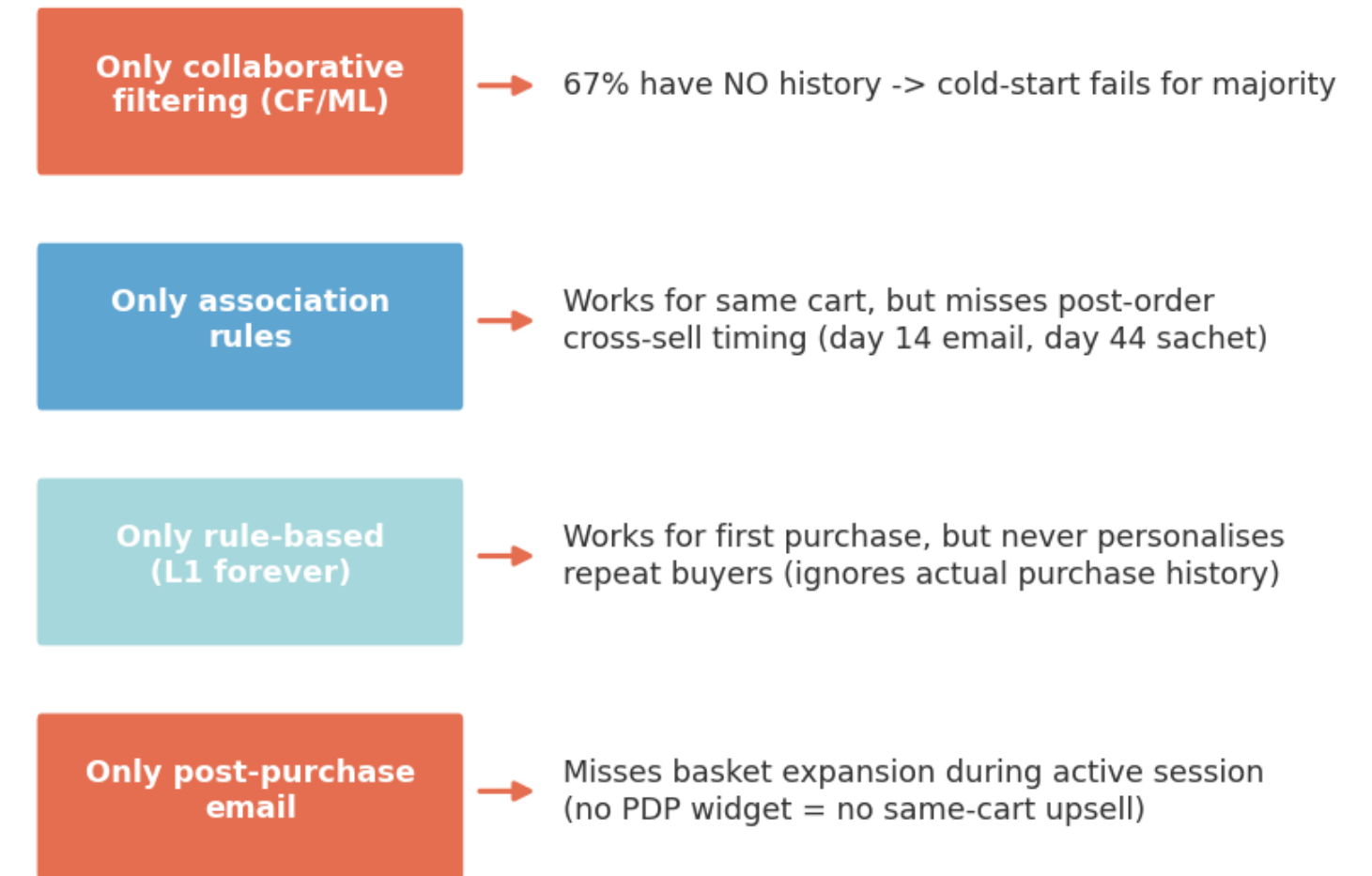
Solution 2 – Recommender System

4 layers breakdown.

Coverage: Why You Need All 4 Layers



What Breaks If You Use Only One Model



KEY INSIGHT:



- **L1 Rule-Based:** Relevant cross-category recommendations from Day 1, **no purchase history needed**
- **L2 Association Rules:** "Frequently Bought Together" suggestions at the **point of purchase**
- **L3 Sequential Timed:** Recommendations delivered **just before the expected reorder window**
- **L4 Item-CF:** Personalized recommendations **for loyal customers (3+ orders)**

Solution 2 – Recommender System



Layer 3 Deep Dive.

First Purchase	Email Day & Flow	Email Recommends	Sample Ships Day	Sample Product	Co-purchase Evidence
Clear Protein 500g	Day 14 (CS-01)	Lean TMT 1kg or Lean Taro 1kg	Day 44 (-10d gap)	Lean 40g single-serve OR Collagen 25g sachet	53% of D1 Clear buyers also buy Lean
Lean Protein 1kg	Day 14 (CS-02)	Clear Peach 500g or Clear White Grape	Day 25 (-10d gap)	Clear 25g sachet OR Collagen 25g sachet	65% of D1 Lean buyers also buy Clear
Collagen Glow 300g	Day 21 (CS-03)	Clear Peach 500g (protein starter)	Day 32 (-10d gap)	Clear 25g sachet OR Lean 40g	30.5% Collagen-first repeat; needs protein
Accessories (Shaker)	Day 7 (CS-04 URGENT)	Lean TMT 1kg (protein trial)	Day 25 (-10d gap)	Clear 25g sachet NOT another shaker	41% Accessories buyers move to Lean
Soy Protein 1kg	Day 14 (CS-01)	Clear Peach 500g (hero gateway)	Day 74 (-10d gap)	Clear or Lean 25g/40g single-serve	Low co-purchase; push hero protein

- Note:**
- **L3 is the most operationally unique layer** – it requires knowing when to fire, what sample to send, and how to route it through fulfilment
 - The timing is not arbitrary (it is calculated from real reorder data from repeat buyers)
 - Push for subscription

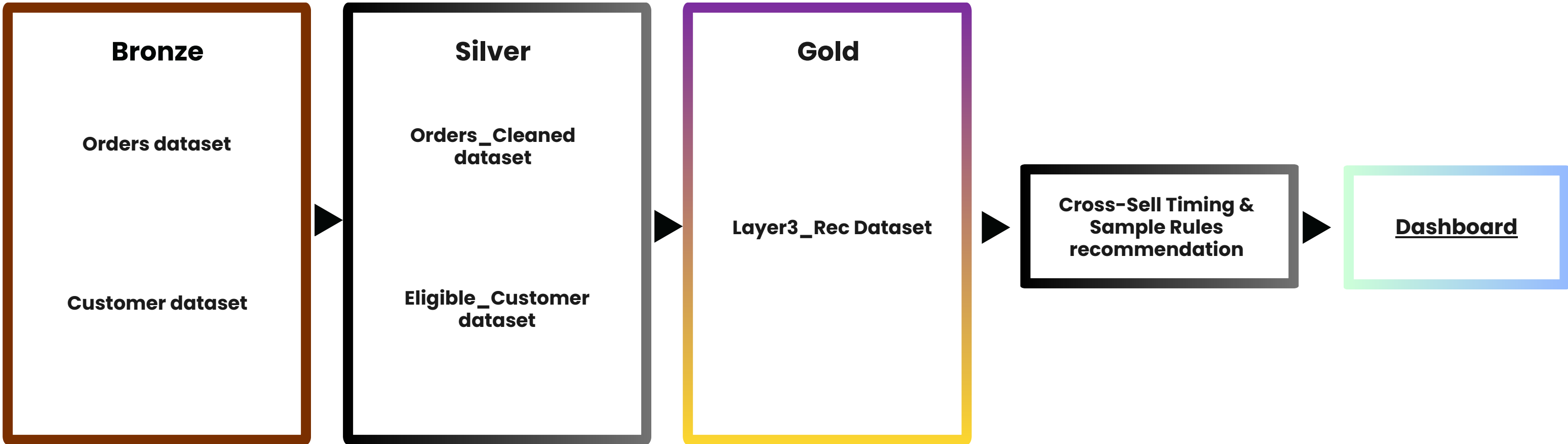
After Order 2 fulfilled: Day 7 email recommends 3rd category | **Order 2 + 48 days:** SUB-01 Subscribe & Save on hero SKU

Solution 2 – Recommender System



Implementation.

PIPELINE FLOW:



Layer 3 Conservative Prize (5% conversion rate assumption)

5% of 3,681 single-cat customers reach 2 categories	+S\$5,152 GP/yr
5% of 3,681 single-cat customers reach 3 categories	+S\$10,693 GP/yr
Subscribe & Save conversion — SUB-01 (689 targets)	+S\$2,262 GP/yr
Acquisition mix fix — stop shaker-led campaigns	+S\$4,389 GP/yr

CONSERVATIVE TOTAL

S\$17 – 30K GP/yr

Conclusion

Insights

- **11.7%** of customers in **Tier 1** → a small but highly valuable group to prioritize for retention
- **42.8%** of customers in **Tiers 2–4** → significant opportunities to increase long-term customer value through targeted interventions.

Solutions

Tier-based Customer Treatment

- Prioritize retention for high-value customers
- Targeted win-back for lapsed customers

4-Layer Recommender System

- Recommend the right products at different stages.
- Encourage cross-selling and subscription adoption

Next Steps & Suggestions

Track onsite event conversion rates

- Encourage customers to create customer accounts during events in exchange for vouchers
- Ensures traceability of event conversion rates by kickstarting online purchase lifecycle

Thank you!



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66% Scale to Window

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Properties

Filters

Scenes

Scene

Sources

Window Capture

Display Capture

Audio Mixer

Desktop Audio

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Scene Transitions

Fade

Duration 300 ms

Controls

Start Streaming

Stop Recording

Start Virtual Camera

Studio Mode

Settings

Exit

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Appendix

Layer 3 — Data Evidence: Every Design Decision is Grounded in the Numbers

Why does the sample ship separately — not in the first order box?

Sachet in first box = ignored

Customer is excited about their new product. The sachet sits unnoticed. By the time they try it, they've already made their reorder decision.

Pre-reorder arrival = maximum influence

Day 44 (for Clear Protein buyers): customer is running low on their protein. They're thinking 'what do I reorder?' — then a Lean TMT sachet arrives. They try it. They add it to their next order.

The sample appears at the decision moment

This is the key insight. The sample doesn't just introduce a product — it shows up at exactly the moment the customer is deciding what to buy next.

How the reorder timing was measured (not assumed)

- 1. For each (SKU, customer): collect all order dates for that SKU.
- 2. Sort dates. Compute day gaps between consecutive orders using .diff().
- 3. Take the MEDIAN gap per customer (handles irregular / bulk buyers).
- 4. Take the MEDIAN across all customers for that SKU.
- 5. Sample dispatch day = median - 10 days.

Why median, not mean? Clear Peach: median=54d vs mean=76d. Lean TMT: median=35d vs mean=69d. Mean is skewed by customers who pause or buy in bulk. Median targets the typical buyer.

SKU	Buyers	Median	Mean	Sample ships
Clear Peach 500g	81	54 days	76 days	Day 44
Clear White Grape	48	54 days	71 days	Day 44
Collagen Glow 300g	40	42 days	50 days	Day 32
Lean TMT 1kg	39	35 days	69 days	Day 25
Creatine 250g	37	66 days	85 days	Day 55

Co-purchase rates — the source of the routing rules

Based on D1 customers (top 10% by profit) — your best customers' actual purchase behaviour.

Bought Clear Protein	-> Lean Protein	53%	D1 Clear buyers also buy Lean
Bought Lean Protein	-> Clear Protein	65%	D1 Lean buyers also buy Clear
Bought Collagen Glow	-> Clear Protein	24%	Attach rate — protein cross-sell
Bought Accessories	-> Lean Protein (URGENT)	41%	Accessories buyers move to protein
Bought Soy Protein	-> Clear Protein	10%	Low co-purchase — push hero gateway SKU

The category ladder prize — what L3 is trying to unlock

Categories ever bought	Customers	Repeat rate	Avg GP
1 category	3,681 (65%)	13%	\$64
2 categories	1,411 (25%)	30%	\$92
3+ categories	602 (11%)	63%	\$223

What each category step means:

+S\$28 avg GP per customer Avg gross profit increase from 1-cat to 2-cat

+17 percentage points Repeat rate increase per category added

+S\$131 avg GP per customer Avg gross profit increase from 2-cat to 3-cat

5% of 3,681 reach 2 categories: 184 custs x +S\$28 = +S\$5,152 GP/yr

5% of 3,681 reach 3 categories: 184 custs x +S\$159 = +S\$10,693 GP/yr